Khashayar Gatmiry

Berkeley, CA | gatmiry@berkeley.edu | gatmiry.github.io | LinkedIn | Google Scholar

Education

UC Berkeley, Simons Institute, Postdoc — August 2025 - Present
- **Advisors:** Peter Bartlett, Emmanuel Abbe

Apple, Part-time Researcher, MLR team — August 2025 - Present
- **Advisors:** Emmanuel Abbe, Samy Bengio

Massachusetts Institute of Technology, PhD in Computer Science — Sept 2019 – August 2025
- **Advisors:** Stefanie Jegelka, Jonathan Adam Kelner

Sharif University of Technology, B.S. in Computer Engineering & Mathematics — Aug 2014, June 2019
- **Mentors:** Manuel Gomez Rodriguez, Abolfazl Motahari

Research Interests & Selected Publications

Machine Learning, Deep Learning, Transformers, Language Models, Diffusion models, Optimization

Selected Publications

- Transformers Learn Two Competing Sorting Algorithms, and Only One Length-Generalizes

  Khashayar Gatmiry, ***Nathan W. Henry***, Chloe H. Su, Shan Chen, Melanie Weber, Sham Kakade, Emmanuel Abbe, Samy Bengio, Jonathan A. Kelner, Peter Bartlett,

  PREPRINT (submitted to ICLR 2027)
- Learn Your Own Thoughts: Abstract Token Curriculum

  Khashayar Gatmiry, Parsa Mirtaheri, Avrajit Ghosh, Jonathan A. Kelner, Holden Lee, Jason D. Lee, Nika Haghtalab, Mikhail Belkin, Emmanuel Abbe, Peter Bartlett,

  PREPRINT, 2026
- Can Looped Transformers Learn to Implement Multi-step Gradient Descent for In-context Learning?

  Khashayar Gatmiry, Nikunj Saunshi, Sashank J. Reddi, Stefanie Jegelka, Sanjiv Kumar

  ICML, 2024
- The Inductive Bias of Flatness Regularization for Deep Matrix Factorization

  Khashayar Gatmiry, Zhiyuan Li, Ching-Yao Chuang, Sashank J. Reddi, Tengyu Ma, Stefanie Jegelka

  NEURIPS, 2023
- What does guidance do? a fine-grained analysis in a simple setting

  Muthu Chidambaram, ***Khashayar Gatmiry***, Sitan Chen, Holden Lee, Jianfeng Lu,

  NEURIPS, 2024
- Learning Mixtures of Gaussians Using Diffusion Models

  Khashayar Gatmiry, Jonathan A. Kelner, Holden Lee

  PREPRINT, 2024

Industry Experience

Research Intern, Microsoft – Redmond, WA — June 2024 – Aug 2024
- **Mentors:** Ronen Eldan, Adil Salim, Yi Zhang
- Worked on generating paragraph semantic embeddings with small language models (SLMs), based on pytorch

Research intern, Google – New York — Oct 2022 – Jan 2023
- **Mentors:** Sashank J. Reddi, Sobhan Miryoosefi
- Trained language models using novel gradient estimates, authored 4 papers in collaboration with my group

Research Visiting Experience

Visiting Researcher, Yale Computer Science – New Heaven, CT July 2023 – Aug 2023

- **Mentor:** Andre Wibisono
- Worked on Hamiltonian Monte Carlo and Gibbs sampler

Visiting Researcher, Georgia Tech College of Computing – Atlanta, GA Aug 2021 – Sep 2021

- **Mentor:** Santosh S. Vempala
- worked on proving isoperimetric inequalities on manifolds, KLS conjecture, and sampling from polytopes Authored 2 publications from projects related to my visit

Research Intern, Max Plank Institute for Software Systems – Kaiserslautern, GER June 2018 – Aug 2018

- **Mentor:** Manuel Gomez Rodriguez
- worked on the network visibility problem and submodularity

Review Service

Reviewer for Neural Information Processing System (NEURIPS), Conference on Learning Representations (ICLR), Fundamentals of Computer Science (FOCS), Symposium on Discrete Algorithms (SODA), Journal of Machine Learning Research (JMLR), Theoretical Computer Science (TCS)

Teaching Service

Teaching Assistant for Introduction to Probability and Statistics (18.05), MIT	Spring 2024
Teaching Assistant for Computation Theory and Automata, Sharif university	Spring 2018
Teaching Assistant for Artificial Intelligence, Sharif university	Spring 2018
Teaching Assistant for Probability and Statistics, Sharif university	Fall 2017
Teaching Assistant for Discrete Structures, Sharif university	Fall 2016
Teaching Math Olympiad in high schools during my undergraduate studies	

Awards

Fellowship from the Institute of Science and Technology Austria (**IST**)	Summer 2018
Outstanding Student Award from Sharif University of Technology	2018
Recipient of the grant for exceptional talents *National Elites Foundation of Iran*	undergraduate studies
Gold Medal, 27th Iranian National Olmypiad in Mathematics	2013
Ranked 4th, International Tournament of Young Mathematicians (**ITYM**), Germany	2014
Silver Medal, International Mathematics Competition (**IMC**), Taiwan	2012
Bronze Medal, International Mathematics Competition (**IMC**), Indonesia	2011
Gold Medal, Asian Inter-cities Teenagers Mathematics Olympiad (**AITMO**), Nepal	2011

Publications

Learn Your Own Thoughts: Abstract Token Curriculum

Khashayar Gatmiry, Parsa Mirtaheri, Avrajit Ghosh, Jonathan A. Kelner, Holden Lee, Jason D. Lee, Nika Haghtalab, Mikhail Belkin, Emmanuel Abbe, Peter Bartlett,

PREPRINT (submitted to ICLR 2027)

Transformers Learn Two Competing Sorting Algorithms, and Only One Length-Generalizes

Khashayar Gatmiry, ***Nathan W. Henry***, Chloe H. Su, Shan Chen, Melanie Weber, Sham Kakade, Emmanuel Abbe, Samy Bengio, Jonathan A. Kelner, Peter Bartlett,

PREPRINT (submitted to ICLR 2027)

RLTL;DR: Self-improvement by Internalizing Self-generated Feedback

Michael Kirchhof, Eleonora Gualdoni, Andrew Szot, ***Khashayar Gatmiry***, Aryo Lotfi, Omar Attia, Abbas Kazerouni, Sanjoy Chowdhury, Alexander T Toshev

PREPRINT (submitted to ICLR 2027)

Online Dense Reinforcement Learning with Planning-Grounding Agent for UI Navigation on Live IOS Apps
Peitong Duan, ***Khashayar Gatmiry***, Jeffrey Nichols, Samy Bengio, Emmanuel Abbe
PREPRINT, 2026

Blackwell-Guided Logit Search (BLS): Optimizing Multi-Turn Jailbreaks with Subliminal Steering
Khashayar Gatmiry, Timothy Ng, Arhaan Garg, Avrajit Ghosh, Emmanuel Abbe, Peter Bartlett, Bin Yu
PREPRINT (submitted to ICLR 2027)

High-accuracy and dimension-free sampling with diffusions
Khashayar Gatmiry, Sitan Chen, Adil Salim,
ICML, 2026

Learning mixtures of gaussians using diffusion models
Khashayar Gatmiry, Jonathan A. Kelner, Holden Lee,
COLT, 2025

On the Role of Depth and Looping for In-Context Learning with Task Diversity
Khashayar Gatmiry, Nikunj Saunshi, Sashank J. Reddi, Stefanie Jegelka, Sanjiv Kumar,
PREPRINT, 2024

Can Looped Transformers Learn to Implement Multi-step Gradient Descent for In-context Learning?
Khashayar Gatmiry, Nikunj Saunshi, Sashank J. Reddi, Stefanie Jegelka, Sanjiv Kumar,
ICML, 2024

Simplicity Bias via Global Convergence of Sharpness Minimization
Khashayar Gatmiry, Zhiyuan Li, Sashank J. Reddi, Stefanie Jegelka
ICML, 2024

What does guidance do? A fine-grained analysis in a simple setting
Muthu Chidambaram, ***Khashayar Gatmiry***, Sitan Chen, Holden Lee, Jianfeng Lu,
NEURIPS, 2024

Adversarial Online Learning with Temporal Feedback Graphs
Khashayar Gatmiry, Jon Schneider, Stefanie Jegelka,
COLT, 2024

Sampling Polytopes with Riemannian HMC: Faster Mixing via the Lewis Weights Barrier
Khashayar Gatmiry, Jonathan A. Kelner, Santosh Vempala,
COLT, 2024

EM for Mixture of Linear Regression with Clustered Data
Amir Reisizadeh, ***Khashayar Gatmiry***, Asuman Ozdaglar,
AISTATS, 2024

Bandit Algorithms for Prophet Inequality and Pandora's Box
Khashayar Gatmiry, Thomas Kesselheim, Sahil Singla, Yifan Wang
SODA, 2024

Rethinking Invariance in In-context Learning
Lizhe Fang, Yifei Wang, ***Khashayar Gatmiry***, Yisen Wang
ICML WORKSHOP, 2024

Projection-free online convex optimization via efficient newton iterations
Khashayar Gatmiry, Zak Mhammedi

NeuRIPS, 2023

The Inductive Bias of Flatness Regularization for Deep Matrix Factorization
Khashayar Gatmiry, Zhiyuan Li, Tengyu Ma, Sashank J. Reddi, Stefanie Jegelka, Ching-Yao Chuang,
NeuRIPS, 2023

Quasi-newton steps for efficient online exp-concave optimization
Khashayar Gatmiry, Zak Mhammedi
COLT, 2023

Quasi-newton steps for efficient online exp-concave optimization
Khashayar Gatmiry, Zak Mhammedi
COLT, 2023

When does Metropolized Hamiltonian Monte Carlo provably outperform Metropolis-adjusted Langevin algorithm?
Yuansi Chen, ***Khashayar Gatmiry***
PREPRINT, 2023

A unified approach to controlling implicit regularization via mirror descent
Haoyuan Sun, ***Khashayar Gatmiry***, Kwangjun Ahn, Navid Azizan
JMLR, 2023

Optimal algorithms for group distributionally robust optimization and beyond
Tasuku Soma, ***Khashayar Gatmiry***, Stefanie Jegelka
PREPRINT, 2022

On the generalization of learning algorithms that do not converge
Nisha Chandramoorthy, Andreas Loukas, ***Khashayar Gatmiry***, Stefanie Jegelka
NeuRIPS, 2022

Adaptive Generalization and Optimization of Three-Layer Neural Networks
Khashayar Gatmiry, Stefanie Jegelka, Jonathan A. Kelner
ICLR, 2022

Convergence of the riemannian langevin algorithm
Khashayar Gatmiry, Santosh S. Vempala
PREPRINT

Testing determinantal point processes
Khashayar Gatmiry, Maryam Aliakbarpour, Stefanie Jegelka
NeuRIPS, 2020

The network visibility problem
Khashayar Gatmiry, Manuel Gomez-Rodriguez
ACM TOIS, 2021